Experiment 5

Compute Measures of Central Tendency and Dispersion Using the State-wise Literacy Rate Dataset in R

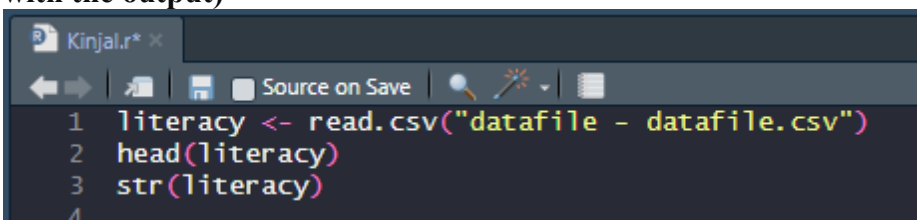
PART B

PART B: TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the ERP or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no ERP access available)

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Class : SE-C	Batch : C2
Date of Experiment:	Date of Submission :
Grade :	

Observation: (Run the programs given above in R studio and paste sceenshots of along with the output)



Observation Table:

```
Console  Terminal  Background Jobs
R 4.6.1 · C:/Users/Admin/Downloads/
> literacy <- read.csv("datafile - datafile.csv")
> head(literacy)
                  STATE.UT      DISTRICT   JAN  FEB  MAR  APR  MAY  JUN  JUL  AUG  SEP  OCT  NOV  DEC ANNUAL JAN.FEB   MAM
1 ANDAMAN And NICOBAR ISLANDS       NICOBAR 107.3 57.9  65.2 117.0 358.5 295.5 285.0 271.9 354.8 326.0 315.2 250.9 2805.2   165.2 540.7
2 ANDAMAN And NICOBAR ISLANDS SOUTH ANDAMAN  43.7 26.0  18.6  90.5 374.4 457.2 421.3 423.1 455.6 301.2 275.8 128.3 3015.7    69.7 483.5
3 ANDAMAN And NICOBAR ISLANDS N & M ANDAMAN  32.7 15.9   8.6  53.4 343.6 503.3 465.4 460.9 454.8 276.1 198.6 100.0 2913.3    48.6 405.6
4           ARUNACHAL PRADESH         LOHIT  42.2 80.8 176.4 358.5 306.4 447.0 660.1 427.8 313.6 167.1  34.1  29.8 3043.8   123.0 841.3
5           ARUNACHAL PRADESH    EAST SIANG  33.3 79.5 105.9 216.5 323.0 738.3 990.9 711.2 568.0 206.9  29.5  31.7 4034.7   112.8 645.4
6           ARUNACHAL PRADESH SUBANSIRI F.D  28.0 48.3  85.3 101.5 140.5 228.4 217.4 182.8 159.8  75.9  20.9  11.6 1300.4    76.3 327.3
    JJAS   OND
1 1207.2 892.1
2 1757.2 705.3
3 1884.4 574.7
4 1848.5 231.0
5 3008.4 268.1
6  788.4 108.4
> str(literacy)
'data.frame':   640 obs. of  19 variables:
 $ STATE.UT: chr  "ANDAMAN And NICOBAR ISLANDS" "ANDAMAN And NICOBAR ISLANDS" "ANDAMAN And NICOBAR ISLANDS" "ARUNACHAL PRADESH" ...
 $ DISTRICT: chr  "NICOBAR" "SOUTH ANDAMAN" "N & M ANDAMAN" "LOHIT" ...
 $ JAN     : num  107.3 43.7 32.7 42.2 33.3 ...
 $ FEB     : num  57.9 26 15.9 80.8 79.5 ...
 $ MAR     : num  65.2 18.6 8.6 176.4 105.9 ...
 $ APR     : num  117 90.5 53.4 358.5 216.5 ...
 $ MAY     : num  358 374 344 306 323 ...
 $ JUN     : num  296 457 503 447 738 ...
 $ JUL     : num  285 421 465 660 991 ...
 $ AUG     : num  272 423 461 428 711 ...
 $ SEP     : num  355 456 455 314 568 ...
 $ OCT     : num  326 301 276 167 207 ...
 $ NOV     : num  315.2 275.8 198.6 34.1 29.5 ...
 $ DEC     : num  250.9 128.3 100 29.8 31.7 ...
 $ ANNUAL  : num  2805 3016 2913 3044 4035 ...
 $ JAN.FEB : num  165.2 69.7 48.6 123 112.8 ...
 $ MAM     : num  541 484 406 841 645 ...
 $ JJAS    : num  1207 1757 1884 1848 3008 ...
 $ OND     : num  892 705 575 231 268 ...
```

Observation : Paste screenshots of the outputs of above programs mentioned

INPUT

Kinjal.r*

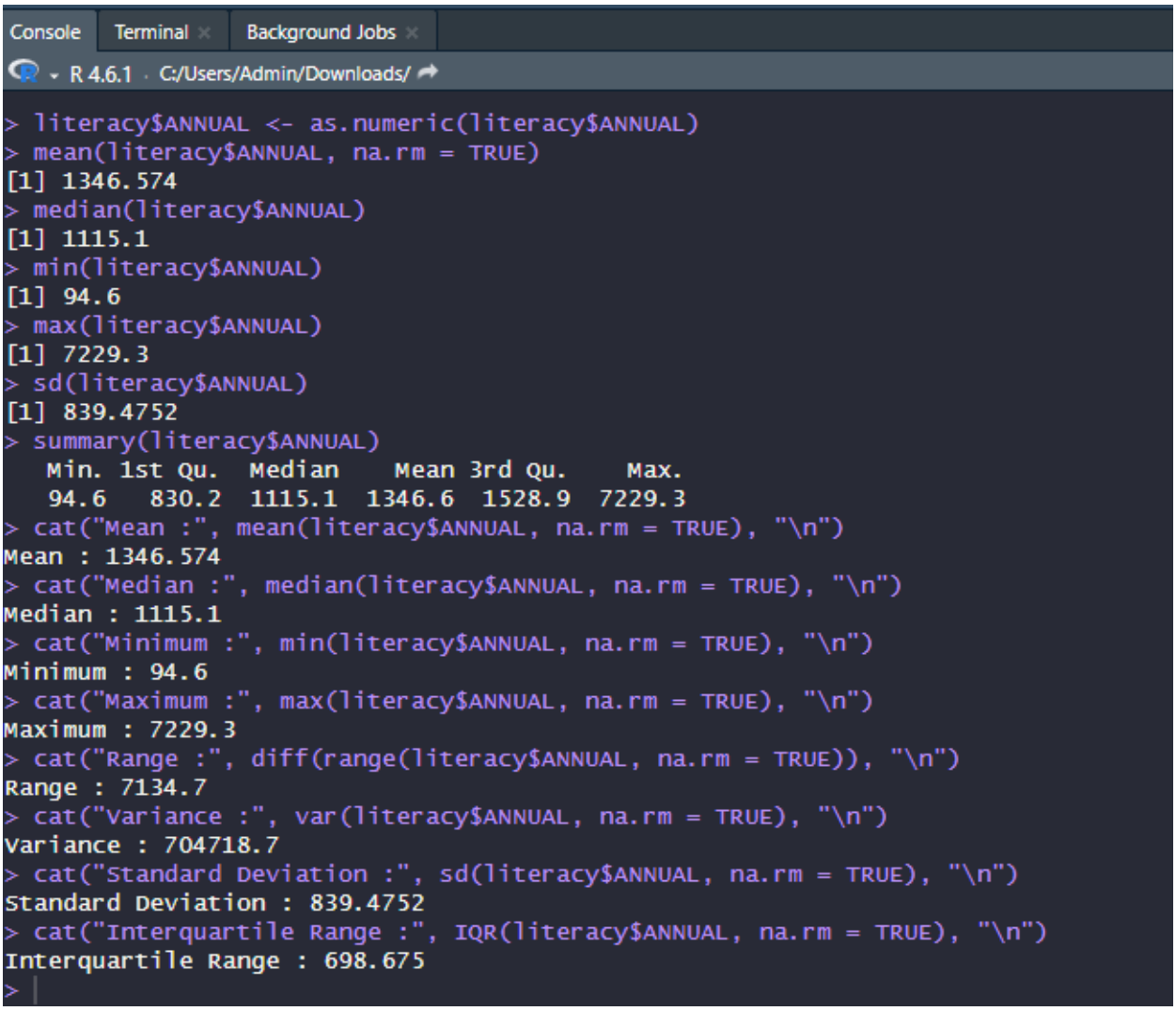
Source on Save

```r
literacy <- read.csv("datafile - datafile.csv")
head(literacy)
str(literacy)


mean(literacy$ANNUAL, na.rm = TRUE)
median(literacy$ANNUAL)
min(literacy$ANNUAL)
max(literacy$ANNUAL)
sd(literacy$ANNUAL)
summary(literacy$ANNUAL)

cat("Mean :", mean(literacy$ANNUAL, na.rm = TRUE), "\n")
cat("Median :", median(literacy$ANNUAL, na.rm = TRUE), "\n")
cat("Minimum :", min(literacy$ANNUAL, na.rm = TRUE), "\n")
cat("Maximum :", max(literacy$ANNUAL, na.rm = TRUE), "\n")
cat("Range :", diff(range(literacy$ANNUAL, na.rm = TRUE)), "\n")
cat("Variance :", var(literacy$ANNUAL, na.rm = TRUE), "\n")
cat("Standard Deviation :", sd(literacy$ANNUAL, na.rm = TRUE), "\n")
cat("Interquartile Range :", IQR(literacy$ANNUAL, na.rm = TRUE), "\n")
```

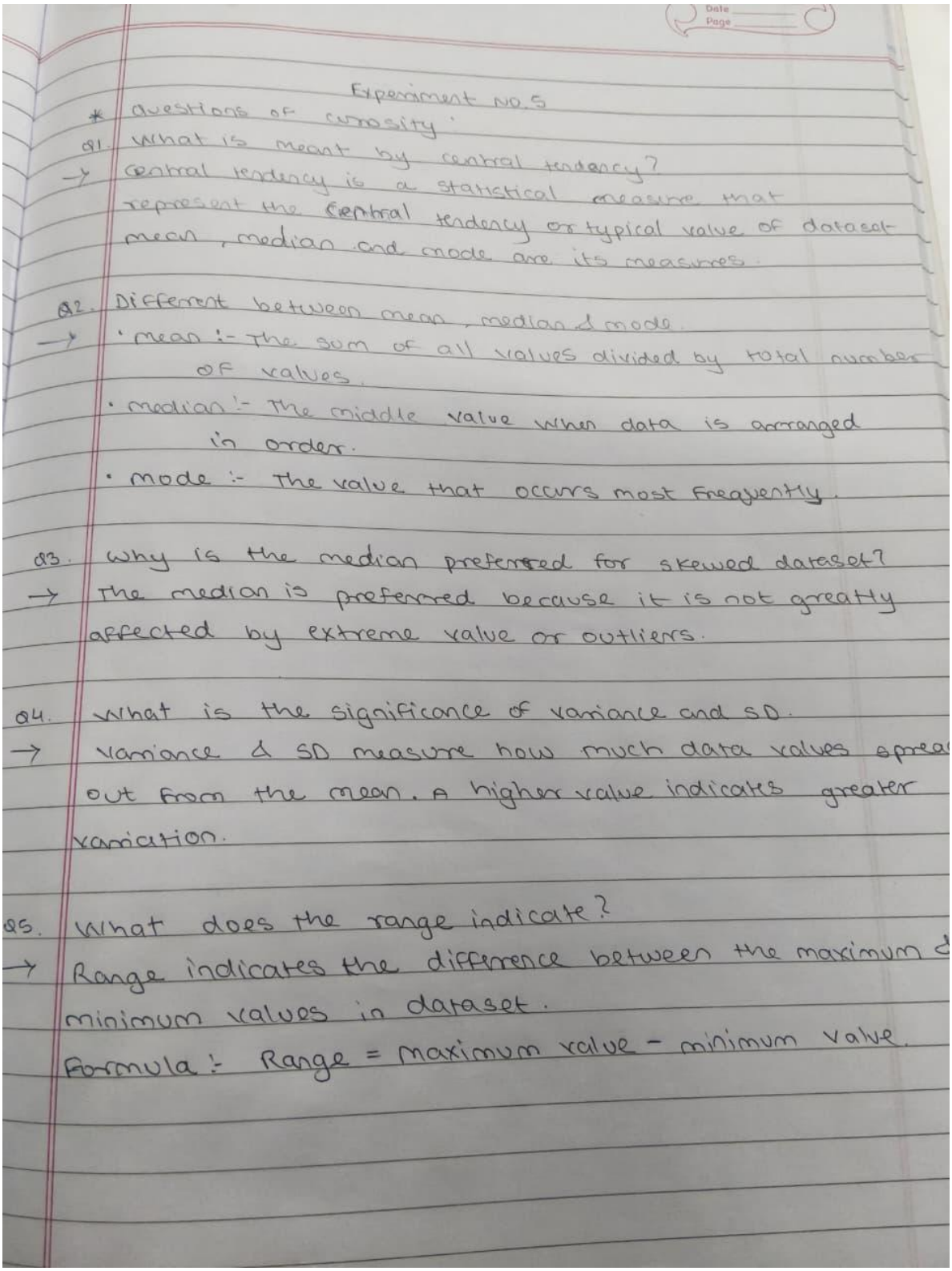
OUTPUTS:



Result: The statistical measures of the ANNUAL literacy data were successfully calculated in R. The mean, median, mode, minimum, maximum, range, variance, standard deviation, quartiles, interquartile range, and summary statistics were generated.

Conclusion: The experiment was completed successfully. R was used to analyze the ANNUAL literacy dataset by calculating measures of central tendency and dispersion, providing a clear summary of the distribution and variability of the literacy values.

Curiosity Questions:



Experiment No. 5

* Questions of curiosity:

Q1. What is meant by central tendency?

→ Central tendency is a statistical measure that represent the central tendency or typical value of dataset- mean, median and mode are its measures.

Q2. Different between mean, median & mode.

→ • mean :- The sum of all values divided by total number of values.

• median :- The middle value when data is arranged in order.

• mode :- The value that occurs most freqyently.

Q3. Why is the median preferred for skewed dataset?

→ The median is preferred because it is not greatly affected by extreme value or outliers.

Q4. What is the significance of variance and SD.

→ Variance & SD measure how much data values spread out from the mean. A higher value indicates greater variation.

Q5. What does the range indicate?

→ Range indicates the difference between the maximum & minimum values in dataset.

Formula :- Range = maximum value - minimum value.

Viva voce Questions:

1. What is meant by central tendency?
2. Differentiate between mean, median, and mode.
3. Why is the median preferred for skewed datasets?
4. What is the significance of variance and standard deviation?
5. What does the range indicate?
6. Explain the importance of the Interquartile Range (IQR).
7. Which R functions are used to calculate mean, median, and standard deviation?
8. Why is preprocessing necessary before statistical analysis?
9. How do outliers affect the mean and standard deviation?
10. How can descriptive statistics help policymakers analyze literacy rates across states?